

Elbow MRI — 30 Sample Cases

A teaching file of 30 worked elbow MRI cases spanning the most common tendon, ligament, cartilage, osseous, and nerve pathologies encountered in clinical practice.

Tendons

Case 1 — Common extensor tendinosis (lateral epicondylitis)

Clinical: 47F recreational tennis player, 4 months of lateral elbow pain worse with gripping.

F: The **common extensor tendon** origin at the lateral epicondyle is thickened and shows intermediate intrasubstance signal on T1 and **mild-to-moderate hyperintensity on fluid-sensitive (PD/T2 fat-sat) sequences**, without a discrete fluid-cleft tear. The tendon measures up to 5 mm in AP thickness. The underlying **lateral epicondyle cortex** is intact; mild adjacent subcortical marrow edema is present. The **LUCL** is intact and well visualized deep to the extensor origin. No fluid-signal gap, no retraction. No joint effusion.

I: Common extensor tendinosis consistent with lateral epicondylitis; no tendon tear.

Case 2 — Common extensor high-grade tear

Clinical: 52M manual laborer, acute worsening of chronic lateral elbow pain after lifting.

F: The **common extensor origin** demonstrates a **high-grade partial-thickness tear**, with a fluid-signal cleft involving approximately 70% of the tendon footprint width and a craniocaudal extent of 9 mm on coronal PD fat-sat images. There is associated **fluid undermining the deep fibers** and mild proximal fiber retraction of 3 mm. Reactive marrow edema is present in the lateral epicondyle. The adjacent **LUCL** shows mild intermediate signal but remains continuous (concurrent low-grade sprain). Small joint effusion.

I: High-grade partial tear of the common extensor tendon with associated lateral epicondyle marrow edema; LUCL intact.

Case 3 — Common flexor tendinosis (medial epicondylitis)

Clinical: 39M golfer, medial elbow pain aggravated by wrist flexion/pronation.

F: The **common flexor-pronator tendon** at the medial epicondyle is thickened with intermediate-to-increased intrasubstance signal on PD fat-sat, without a discrete fluid cleft or full-thickness defect. The deep **anterior band of the UCL** immediately deep to the flexor origin is intact and normal in caliber. No marrow edema beyond mild reactive change at the medial epicondyle. The **ulnar nerve** in the cubital tunnel is normal in caliber and signal.

I: Common flexor-pronator tendinosis (medial epicondylitis); UCL and ulnar nerve intact.

Case 4 — Distal biceps partial tear

Clinical: 44M, antecubital pain after a single heavy deadlift, weak supination.

F: The **distal biceps tendon** demonstrates a **partial-thickness tear** involving predominantly the deep/radial fibers at the **radial tuberosity** insertion, with intrasubstance and surrounding fluid signal on axial and sagittal FABS-view images. Approximately 50% of the footprint remains attached. The tendon is intact in continuity without retraction. There is mild surrounding edema and trace **bicipitoradial bursa** fluid. The **lacertus fibrosus** is intact.

I: Partial-thickness tear of the distal biceps tendon at the radial tuberosity insertion; tendon in continuity, no retraction.

Case 5 — Distal biceps complete rupture with retraction

Clinical: 53M, sudden antecubital "pop" during eccentric lifting, ecchymosis and Popeye-type deformity.

F: There is a **full-thickness complete rupture of the distal biceps tendon** at the radial tuberosity, with a fluid-filled gap and **proximal retraction of the tendon stump to approximately 4.5 cm above the joint line**. The surrounding hematoma and edema track along the muscle. The **lacertus fibrosus is torn**, accounting for the degree of retraction. The brachialis is intact. Marked **bicipitoradial bursa** distension with hemorrhagic fluid.

I: Complete distal biceps tendon rupture with 4.5 cm retraction and torn lacertus fibrosus; surgically relevant retraction.

Case 6 — Distal triceps tear

Clinical: 49M, posterior elbow pain and weak extension after a fall onto an outstretched arm.

F: The **distal triceps tendon** shows a **partial-thickness tear** at the olecranon insertion involving the superficial fibers, with a fluid-signal gap measuring 6 mm craniocaudally and an intact deep (medial head) component on sagittal PD fat-sat. There is no significant retraction. Adjacent **olecranon** marrow edema and a small **olecranon bursa** effusion are noted. No avulsed osseous fragment.

I: Partial-thickness distal triceps tendon tear at the olecranon insertion with intact deep fibers; no significant retraction.

Case 7 — Bicipitoradial (cubital) bursitis

Clinical: 58F, antecubital fullness and discomfort with forearm rotation.

F: A well-defined fluid-signal collection surrounds the distal biceps tendon at the radial tuberosity, consistent with a distended **bicipitoradial bursa** measuring 3.2 x 1.4 x 2.6 cm, with thin synovial-type enhancement pattern and mild internal septation. The **distal biceps tendon** itself is intact and normal in caliber. Mild mass effect on the adjacent **radial/PIN** region without nerve signal abnormality.

I: Bicipitoradial bursitis with an intact distal biceps tendon; mild mass effect on adjacent soft tissues.

Ligaments & Instability

Case 8 — UCL anterior band partial tear (T-sign)

Clinical: 21M collegiate baseball pitcher, medial elbow pain late in the throwing cycle.

F: The **anterior bundle of the ulnar collateral ligament** demonstrates a **partial undersurface tear at the sublime tubercle**, with contrast/fluid tracking distally between the ligament and the medial coronoid margin producing the characteristic **"T-sign"** on coronal images. The proximal ligament is intact. Mild surrounding edema; the **common flexor origin** shows associated low-grade tendinosis. No full-thickness gap.

I: Partial undersurface (distal) tear of the UCL anterior bundle with T-sign; consistent with valgus overload in a thrower.

Case 9 — UCL complete tear

Clinical: 26M javelin thrower, acute medial pain with a "pop," sense of instability.

F: The **anterior bundle of the UCL** is **completely disrupted**, with a full-thickness fluid-signal gap at the mid-substance and lax, wavy ligament fibers. There is extensive surrounding edema and hemorrhage in the medial soft tissues. Associated strain of the **flexor-pronator mass** and reactive marrow edema at the **sublime tubercle**. Mild valgus widening of the medial ulnohumeral joint.

I: Complete mid-substance tear of the UCL anterior bundle with flexor-pronator strain; surgically relevant in a throwing athlete.

Case 10 — LUCL tear with posterolateral rotatory instability

Clinical: 34M, recurrent clicking and giving-way after a prior elbow dislocation.

F: The **lateral ulnar collateral ligament (LUCL)** is **avulsed from its humeral origin** at the lateral epicondyle with a fluid-signal gap and proximal fiber laxity on coronal PD fat-sat. The remaining **lateral collateral ligament complex** is attenuated. Findings are consistent with the substrate for **posterolateral rotatory instability (PLRI)**. Associated common extensor origin tendinosis and a small posterolateral capsular recess effusion.

I: LUCL humeral-sided avulsion consistent with posterolateral rotatory instability; correlate with lateral pivot-shift testing.

Case 11 — Radial (lateral) collateral complex sprain

Clinical: 30F, lateral elbow pain after a varus stress injury during a fall.

F: The **radial collateral ligament** proper shows intermediate-to-high signal with periligamentous edema but maintained continuity, consistent with a **grade II sprain**. The **LUCL** and **annular ligament** are intact. No osseous avulsion at the lateral epicondyle. Small lateral capsular effusion; common extensor origin intact.

I: Grade II radial collateral ligament sprain with intact LUCL and annular ligament; no instability features.

Case 12 — Valgus extension overload (thrower's elbow)

Clinical: 24M professional pitcher, posteromedial pain at ball release, loss of terminal extension.

F: There is **posteromedial olecranon osteophytosis** (a "kissing" osteophyte) with subcortical marrow edema at the **posteromedial olecranon** and reciprocal changes at the **olecranon fossa**, consistent with **valgus extension overload**. A small posteromedial **intra-articular loose body** (4 mm) is present in the olecranon fossa. The **UCL anterior bundle** shows chronic thickening with low-grade partial tearing. Posterior joint effusion.

I: Valgus extension overload syndrome with posteromedial olecranon osteophyte, chondral wear, loose body, and chronic UCL changes.

Case 13 — Capitellar osteochondritis dissecans (stable)

Clinical: 14M competitive gymnast, lateral elbow pain and stiffness, no locking.

F: A focal subchondral lesion is present in the **anterolateral capitellum** measuring 9 x 7 mm, with mildly hypointense subchondral bone on T1 and **no high T2 signal or fluid undermining the fragment**. The **overlying articular cartilage is intact and contiguous** on PD fat-sat. No displaced fragment or loose body. Trace joint effusion.

I: Stable capitellar osteochondritis dissecans with intact overlying cartilage and no fragment instability.

Case 14 — Capitellar OCD (unstable)

Clinical: 16M baseball pitcher, lateral pain with mechanical catching.

F: The **capitellar OCD lesion** measures 12 x 10 mm with **high T2 fluid signal completely encircling the osteochondral fragment**, an associated overlying cartilage defect, and a small displaced **intra-articular loose body** (5 mm) in the anterior recess. Subchondral cyst formation is present. Moderate joint effusion.

I: Unstable capitellar osteochondritis dissecans with fluid undermining the fragment and a displaced loose body.

Case 15 — Panner disease

Clinical: 8M, dominant-arm lateral elbow ache, no antecedent trauma.

F: The **capitellar ossification center** shows diffuse low T1 signal with patchy fluid-sensitive hyperintensity and mild fragmentation/flattening, **without a discrete dissecting fragment, cartilage defect, or loose body**. The articular surface is preserved. No effusion of significance.

I: Panner disease (osteochondrosis of the capitellar ossification center) in a young child; no fragment instability — favorable for spontaneous recovery.

Case 16 — Radial head fracture with effusion

Clinical: 41F, fall on outstretched hand, lateral elbow pain and limited rotation; radiographs equivocal.

F: A **nondisplaced fracture of the radial head** is present, seen as a linear low-signal line on T1 with surrounding marrow edema on PD fat-sat, without articular step-off (<1 mm). There is a **moderate joint effusion with a positive posterior fat pad sign** and a lipohemarthrosis fluid-fluid level in the olecranon recess. Collateral ligaments intact.

I: Nondisplaced radial head fracture with lipohemarthrosis and positive posterior fat pad sign.

Case 17 — Olecranon stress reaction

Clinical: 23M throwing athlete, deep posterior elbow ache with repetitive extension.

F: There is **subcortical marrow edema in the posteromedial olecranon** with a faint linear low-signal stress line and adjacent periosteal edema, consistent with an **olecranon stress reaction/early stress fracture**. The **coronoid process** tip shows mild reactive edema. No discrete cortical break or displacement. Small posterior effusion; UCL chronically thickened.

I: Olecranon stress reaction (early stress fracture) related to valgus extension overload; correlate with activity modification.

Case 18 — Olecranon bursitis

Clinical: 55M, painless then tender posterior elbow swelling; works leaning on elbows.

F: A **distended olecranon bursa** is present superficial to the olecranon, measuring 5.1 x 2.0 x 4.3 cm, containing fluid with mild internal septations and synovial thickening; no definite rim enhancement to suggest infection, but clinical correlation is advised. The **triceps insertion** and underlying olecranon cortex are intact. No intra-articular communication.

I: Olecranon bursitis; clinically correlate to exclude septic bursitis if erythema/fever present.

Nerves

Case 19 — Cubital tunnel syndrome / ulnar neuritis

Clinical: 46F, medial elbow pain with 4th/5th digit numbness and positive Tinel's at the cubital tunnel.

F: The **ulnar nerve** within the **cubital tunnel** is enlarged and hyperintense on PD fat-sat, with a cross-sectional caliber increase relative to the more proximal segment and loss of normal fascicular architecture. The overlying **cubital tunnel retinaculum (Osborne's ligament)** is intact, and the nerve remains located in the groove. No discrete mass or ganglion. Mild adjacent edema.

I: Ulnar neuritis at the cubital tunnel consistent with cubital tunnel syndrome; no compressive mass.

Case 20 — Ulnar nerve subluxation

Clinical: 33M, snapping sensation at the medial elbow with flexion, intermittent ulnar paresthesias.

F: On dynamic/flexion-positioned axial images the **ulnar nerve subluxes anteriorly over the medial epicondyle**, lying anterior to the epicondylar apex rather than within the retrocondylar groove. The nerve shows mild enlargement and increased T2 signal consistent with **friction neuritis**. A **prominent/anomalous medial head of the triceps (anconeus epitrochlearis variant consideration)** is noted. No fixed compressive lesion.

I: Anterior subluxation of the ulnar nerve over the medial epicondyle with friction neuritis; accounts for snapping symptoms.

Case 21 — Posterior interosseous nerve / radial tunnel syndrome

Clinical: 38F, lateral forearm pain distal to the epicondyle, worse with resisted supination; no fixed weakness.

F: The **posterior interosseous nerve (PIN)** demonstrates increased T2 signal as it passes through the **arcade of Frohse** at the proximal supinator, with mild denervation-pattern edema in the **supinator and extensor digitorum** muscles. No mass or ganglion compressing the nerve. The **radial tunnel** soft tissues are otherwise unremarkable.

I: PIN irritation at the arcade of Frohse with early supinator/extensor denervation edema; consistent with radial tunnel syndrome.

Case 22 — Pronator / median nerve syndrome

Clinical: 42M, volar forearm aching and proximal median paresthesias aggravated by repetitive pronation.

F: The **median nerve** shows increased caliber and T2 signal as it passes between the **two heads of the pronator teres**, with mild perineural edema. Subtle denervation-type edema is noted in the **flexor digitorum superficialis** muscle belly. No bifid nerve, ganglion, or fibrous band mass is demonstrated. A bony **supracondylar process/ligament of Struthers** is not identified.

I: Median nerve irritation at the pronator teres consistent with pronator syndrome; no structural compressive lesion identified.

Other

Case 23 — Joint effusion with posterior fat pad sign

Clinical: 29M, post-traumatic elbow pain and swelling; occult fracture suspected.

F: A **moderate elbow joint effusion** distends the anterior and posterior recesses, producing a **positive posterior fat pad sign** and elevation of the anterior (sail) fat pad on sagittal images. No definite intra-articular fracture line is identified on the MRI; the radial head, coronoid, and capitellum show no marrow edema. Collateral ligaments intact.

I: Moderate joint effusion with positive posterior fat pad sign; no MRI-evident fracture, but correlate clinically given the effusion.

Case 24 — Intra-articular loose bodies

Clinical: 50M with prior trauma, intermittent locking and catching.

F: Multiple **intra-articular osteochondral loose bodies** are identified, the largest measuring 8 mm within the **coronoid fossa**, with additional smaller fragments in the **olecranon fossa** and posterior recess. Background changes of **osteoarthritis** include marginal osteophytes and capitellar chondral thinning. Moderate joint effusion outlines the bodies.

I: Multiple intra-articular loose bodies with underlying osteoarthritis; correlate for mechanical locking symptoms.

Case 25 — Synovitis

Clinical: 44F with inflammatory arthropathy, diffuse elbow pain and morning stiffness.

F: There is **diffuse synovial thickening** lining the joint with intermediate signal on PD and avid enhancement (frond-like pattern) on post-contrast images, accompanied by a joint effusion. Small **marginal erosions** are noted at the radial head and coronoid bare areas. No discrete loose body. Mild adjacent soft tissue edema.

I: Inflammatory synovitis with early marginal erosions; pattern compatible with an inflammatory arthropathy such as rheumatoid arthritis.

Case 26 — Brachialis musculotendinous strain

Clinical: 27M climber, acute anterior elbow pain after a maximal pull, painful flexion.

F: The **brachialis muscle** shows feathery intramuscular edema on PD fat-sat at the distal **musculotendinous junction**, consistent with a **grade I–II strain**, without a discrete fiber gap or significant hematoma. The **distal biceps tendon** is intact. No osseous avulsion at the coronoid (brachialis insertion). Trace surrounding edema.

I: Grade I–II brachialis musculotendinous strain; distal biceps intact.

Case 27 — Triceps musculotendinous strain

Clinical: 31M weightlifter, posterior arm pain after heavy bench press lockout.

F: Feathery edema is present within the **medial and long heads of the triceps** at the distal musculotendinous junction on fluid-sensitive sequences, consistent with a **low-grade muscle strain**, without a discrete tendon tear or retraction. The **olecranon insertion** is intact. No olecranon bursal fluid.

I: Low-grade triceps musculotendinous strain with intact olecranon insertion.

Case 28 — Pediatric medial epicondyle avulsion (Little League elbow)

Clinical: 12M baseball pitcher, acute medial pain after a hard throw.

F: There is **widening and edema of the medial epicondyle apophysis** with an avulsed apophyseal fragment displaced approximately 4 mm on coronal images, and surrounding marrow and soft tissue edema. The **common flexor-pronator origin** is attached to the avulsed fragment. The **UCL** is intact at its attachment to the fragment. No intra-articular fragment incarceration.

I: Acute medial epicondyle apophyseal avulsion (Little League elbow) with 4 mm displacement; flexor-pronator and UCL attached to the fragment.

Case 29 — Pediatric supracondylar fracture

Clinical: 6F, fall onto outstretched hand, painful swollen elbow; radiographs with subtle findings.

F: A **minimally displaced supracondylar fracture** of the distal humerus is seen as a transverse marrow-edema-associated low-signal line with mild posterior angulation, accompanied by a **large joint effusion and positive posterior fat pad sign**. The anterior humeral line is maintained. No frank displacement; neurovascular structures grossly unremarkable on MRI.

I: Minimally displaced supracondylar humerus fracture with joint effusion and posterior fat pad sign; correlate with the anterior humeral line on radiographs.

Case 30 — Elbow osteoarthritis (primary/post-traumatic)

Clinical: 61M former manual laborer, chronic stiffness and loss of terminal extension.

F: There is **diffuse cartilage thinning** across the ulnohumeral and radiocapitellar compartments with **marginal osteophytes** at the coronoid, olecranon, and radial head, subchondral cysts, and reactive marrow edema. A few small **intra-articular loose bodies** are present in the coronoid and olecranon fossae. Moderate joint effusion with mild synovial thickening.

I: Degenerative elbow osteoarthritis with osteophytosis, chondral loss, and loose bodies; findings explain loss of terminal extension.